

University of Naples Federico II
Department of Electrical Engineering and Information Technology
Field and Service Robotics (FSR) – a.y. 2024/2025

Homework n.4

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1 Point

The underwater robotic system can be seen as aerial robots, but instead of considering the spatial body moving in the air, it will be considered the spatial body moving in a fluid and so then, instead of aerodynamic effects, now the hydrodynamic effects have to be taken into account.

These hydrodynamic effects include the current effect, the damping effect and take into account the added mass and inertia effect that must be considered when a rigid body is moving in a fluid because the particle surrounding the object are accelerated with the body itself and produce a force not negligible as the aerial robot case.

Another important hydrodynamic effect that must be take into account is the one that include the so-called restoring forces which are the gravity and the buoyancy. Namely, the buoyancy effect is a restoring force which persist even without any accelerations, thus in stationary condition. As the gravitational force, this effect arise not thanks to the relative movement between the body and the liquid, but just because it is immerse in it.

Starting from these considerations, two important facts can be analyzed. The first thing is that the buoyancy is function of the volume of the body immersed in the liquid, and the density of the liquid itself as can be seen from the following buoyancy formula:

$$b = \rho \Delta \|\bar{g}\|$$

where:

- the density of the water $\rho \in \mathbb{R}$
- the volume of the body $\Delta \in \mathbb{R}$
- the gravity vecotr $\bar{g} = [0 \ 0 \ g]^T \in \mathbb{R}^3$ whit $g \in \mathbb{R}$ the gravity acceleration

The second thing is that the buoyancy effect acts at the center of buoyancy $r_b^b \in \mathbb{R}^3$ which is a different center with respect to the gravity force, which acts at the center of mass $r_c^b \in \mathbb{R}^3$. Starting from this consideration, the buoyancy force can be written as following:

$$f_b^b = -R_b^T \begin{bmatrix} 0 \\ 0 \\ b \end{bmatrix} = -R_b^T \begin{bmatrix} 0 \\ 0 \\ \rho \Delta g \end{bmatrix}$$

This means that the buoyancy force is a force applied in a point different from the origin of the body frame and this fact can be seen in the presence of the rotation matrix R_b^T in the previous expression.

2 Point

- a. FALSE. The added mass effect arises when a rigid body moves through a fluid and depends on what the mass is doing in that moment and so is something related to the inertia, namely, much faster the body is moving, much inertia will have. The movement of the body causes the acceleration of the particle fluid around it and, as consequence, a reaction force appears. This force will be equal in magnitude and opposite in direction. Starting from this consideration it is clear that the added mass effect does not mean that one simply add extra mass to the system like considering heavier load, indeed, this effect depends on the geometry of the mobile robot. Moreover the mass matrix associated to it, is not necessarily positive definite, it becomes only under some assumption such as an ideal fluid, no currents, low-speed velocity, etc.
- b. TRUE. For underwater robots, it is necessary to consider the effect of additional mass. In fact, in the case of aerial robots, the density of air is much lower than the density of the body and is not comparable, while for underwater robots, the density of water is much greater than the density of the body and is comparable. If one consider that during underwater movement, the robot displaces the surrounding water, which provides an additional force due to its inertia, this force must be taken into account when modeling the robot's dynamics.
- c. TRUE. The damping effect has an important role in the stability analysis, indeed, it helps to reduce oscillations and dissipate energy. In aerial and underwater robotics, it comes from the viscosity of the surrounding fluid and it can be considered as the sum of two forces, namely, the drag and lift forces on the body. If only quadratic damping terms are considered, grouped in a positive definite matrix with constant coefficients, these forces are applied at the center of mass, are related to the velocity of the body, and are useful for stability analysis using Lyapunov's theory.
- d. FALSE. The ocean current represents one of the main disturbances problem for underwater robots and therefore is usually assumed constant and irrotational but, in the analysis of underwater robot control, it is best expressed with respect to the world frame and not the body frame because if expressed in the body frame, the ocean current changes continuously as the underwater robot's pose changes, while with a more proper notation in the world frame it can be seen as a constant vector v_c . The effect of the ocean current is added into the dynamic model of the robot by accounting for the relative velocity in the body fixed frame v_r thanks to the rotation matrix R_b^T .

$$v_c = \begin{bmatrix} v_{c,x} \\ v_{c,y} \\ v_{c,z} \\ 0 \\ 0 \\ 0 \end{bmatrix} \in \mathbb{R}^6, \quad v_r = \begin{bmatrix} \dot{p}_b^b \\ \omega_b^b \end{bmatrix} - R_b^T v_c$$

3 Point

Starting from the quadratic problem, inside the `MAIN.m` file, which is located within the `quadruped_simulation.zip` file, in the following form:

$$\begin{aligned} \min \quad & \frac{1}{2}x^\top Hx + g^\top x \\ \text{s.t.} \quad & A_{\text{eq}} x = b_{\text{eq}} \\ & A_{\text{ineq}} x \leq b_{\text{ineq}} \end{aligned}$$

where:

- A_{eq} is a sparse matrix of size (p, n) where p is number of equality constraints
- b_{eq} is a dense column vector of size p
- A_{ineq} is a sparse matrix of size (m, n) where m is the number of inequality constraints
- b_{ineq} is a dense column vector of size m

The quadratic function has been implemented using the QP solver `qpSWIFT` as following where the result of the QP problem have been stored in a variable called `zval` in order to be used during the code compilation

```
[zval, basic_info, adv_info] = qpSWIFT(sparse(H), g, sparse(Aeq), beq,  
                                       sparse(Aineq), bineq);
```

After the computation of the quadratic problem solver, it is possible to perform an analyses on the results. The analyses will be carried out by comparing two different configurations setup for each gait scenario with the aim of checking the results in terms of trajectory tracking, sudden oscillations, and suitable values in terms of robot's position, velocity, angular velocity and, z-component of the ground reaction forces.

Following this approach, the simulation results presented below will show a comparison between the plots of the robot using the basic parameters provided by the `MAIN.m` and `get_params.m` file and the plots of the robot by changing the desired velocity v_d , the mass of the robot $mass$, and the friction coefficient μ . In order to have a complete overview of what happens when each parameter is changed, it has been chosen to vary each parameter first for a smaller value and then for a larger value.

The following is a link to the Github repository where it is possible to view all the simulations that will be discussed.

https://github.com/marcomaffeo/FSR_HW4.git

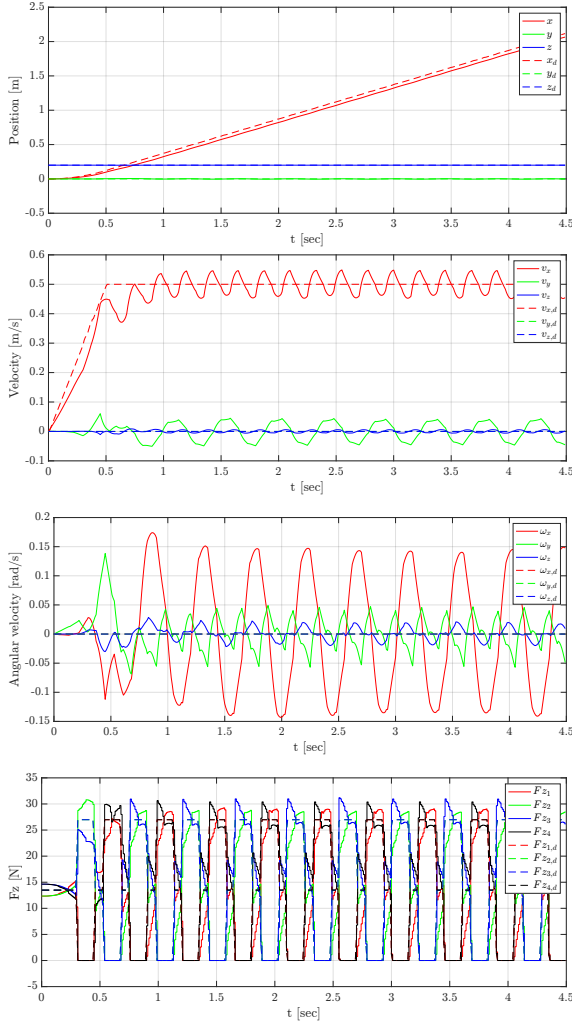


Figure 1: Trot gait configuration results with $v_d = 0.5 \text{ m/s}$, mass = 5.5 kg, $\mu = 1.0$

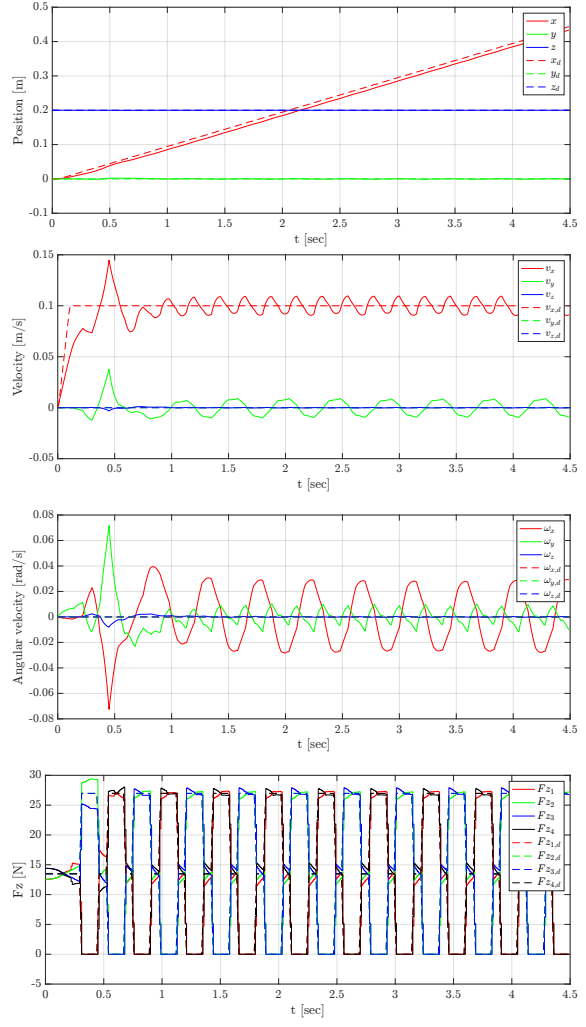


Figure 2: Trot gait configuration results with $v_d = 0.1 \text{ m/s}$, mass = 5.5 kg, $\mu = 1.0$

The first comparison regards the Trot gait configuration. It is possible to see, on the left side (Figure 1), the configuration with the basic parameter values provided by the file, and on the right side (Figure 2), the configuration with adjusted parameters. Namely, the desired velocity has been reduced from 0.5 m/s to 0.1 m/s.

Looking at the position plots, it is possible to see that the tracking and behavior are similar in both cases. However, in the second case, the robot covers less distance—approximately 0.45 m compared to 2.1 m. This is due to the decrease in the desired velocity.

From the velocity plots, it is possible to observe that the desired x -component of the velocity now remains constant at 0.1 m/s instead of 0.5 m/s, while the other component performs almost in the same way. Looking at the angular velocity plot, it is possible to see that, as happened with the linear velocity, there is a decrease in the magnitude of the angular velocity, while the tracking and behavior remain the same. Lastly, looking at the z -component of the ground reaction force plots, it is possible to see that the magnitude is quite similar in both cases. However, in both cases, the tracking of the second component is not particularly accurate, as some spikes are present.

In conclusion, the decrease in desired velocity has no significant effect in terms of improving or worsening the overall performance.

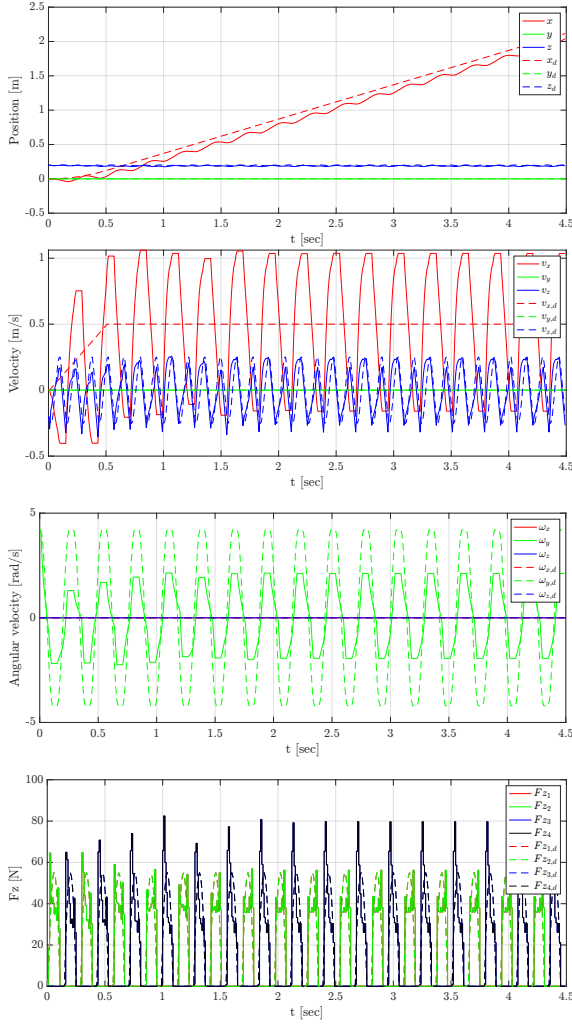


Figure 3: Bound gait configuration results with $v_d = 0.5 \text{ m/s}$, mass = 5.5 kg, $\mu = 1.0$

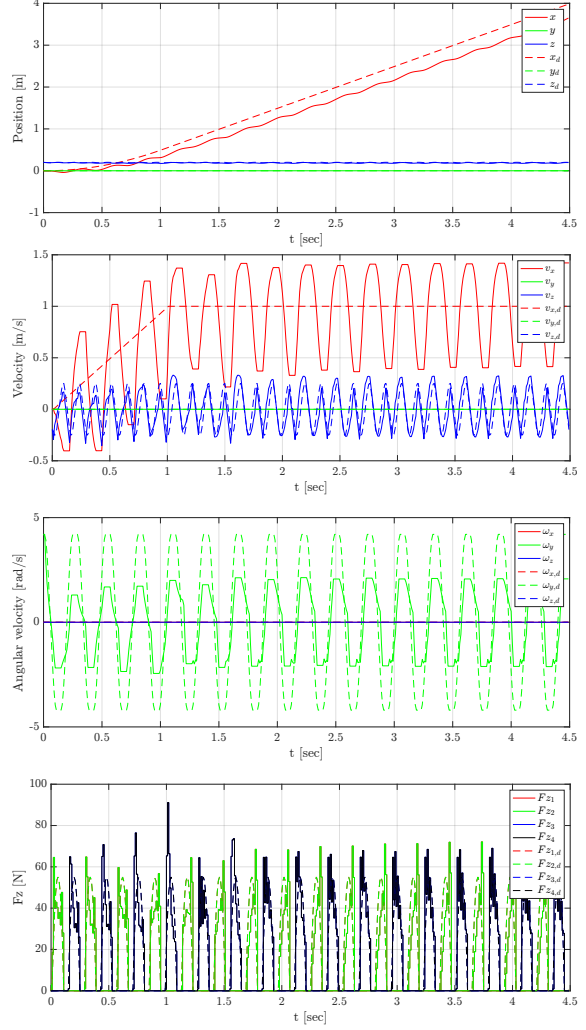


Figure 4: Bound gait configuration results with $v_d = 1.0 \text{ m/s}$, mass = 5.5 kg, $\mu = 1.0$

The second comparison regards the Bound gait configuration. It is possible to see, on the left side (Figure 3), the configuration with the basic parameter values provided by the file, and on the right side (Figure 4), the configuration with adjusted parameters. Namely, the desired velocity has been increased from 0.5 m/s to 1.0 m/s.

Looking at the position plots, it is possible to see that the tracking and behavior are similar in both cases. However, in the second case, the robot covers more distance—approximately 3.5 m compared to 2.1 m. This is due to the increase in the desired velocity.

From the velocity plots, it is possible to observe the x -component of the velocity, where the width of the trapezoidal profile has been slightly reduced, allowing for better tracking, while the other component performs almost in the same way.

Looking at the angular velocity plot, the tracking and behavior remain the same, and the magnitude is also unchanged.

Lastly, looking at the z -component of the ground reaction force plots, it is possible to see that the magnitude is quite similar in both cases.

In conclusion, the increase in desired velocity results in a slight improvement in velocity tracking for the x -component.

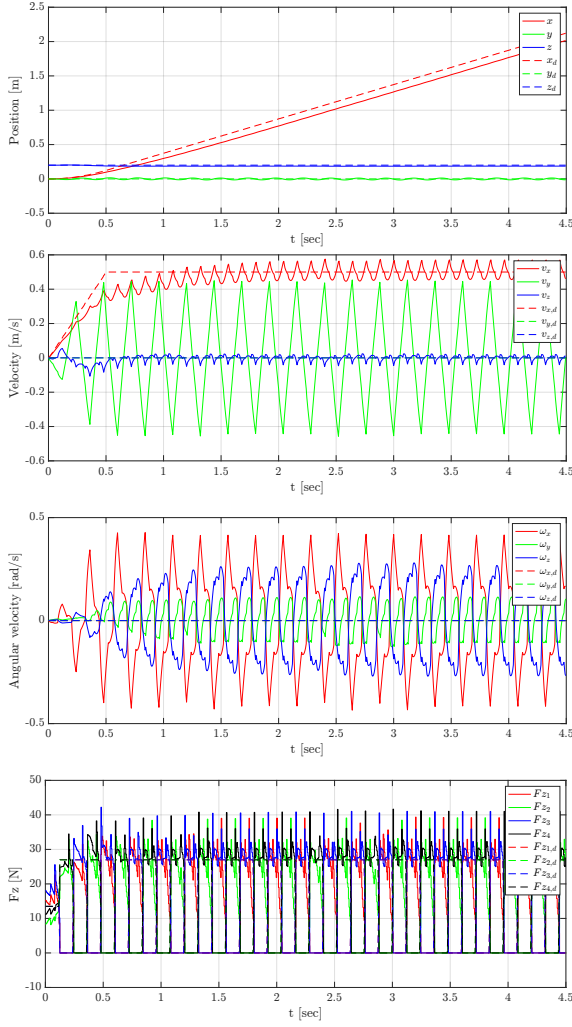


Figure 5: Pacing gait configuration results with $v_d = 0.5 \text{ m/s}$, $\text{mass} = 5.5 \text{ kg}$, $\mu = 1.0$

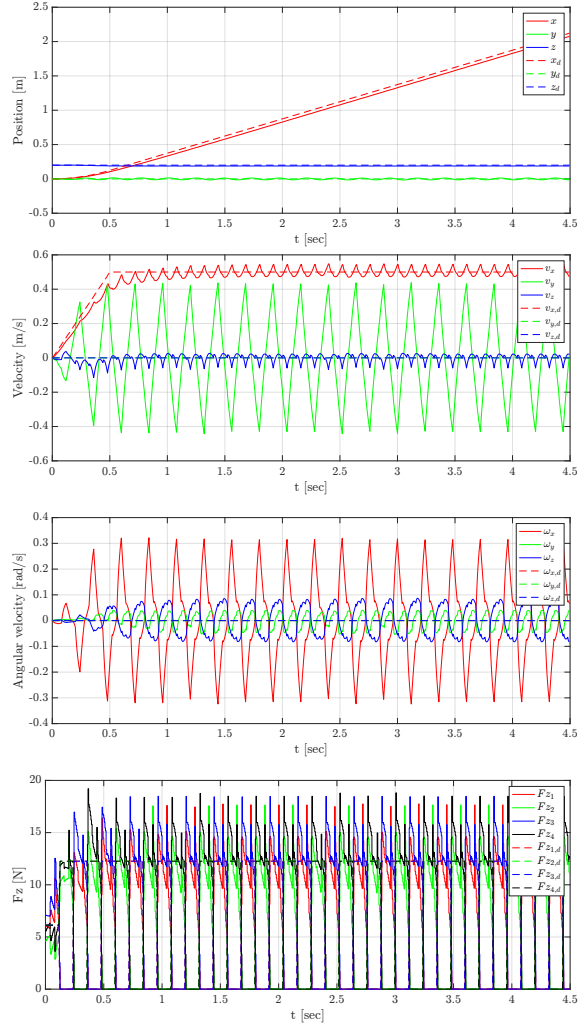


Figure 6: Pacing gait configuration results with $v_d = 0.5 \text{ m/s}$, $\text{mass} = 2.5 \text{ kg}$, $\mu = 1.0$

The third comparison regards the Pacing gait configuration. It is possible to see, on the left side (Figure 5), the configuration with the basic parameter values provided by the file, and on the right side (Figure 6), the configuration with adjusted parameters. Namely, the robot mass has been decreased from 5.5 kg to 2.5 kg.

Looking at the position plots, it is possible to see that the tracking and behavior are similar in both cases. However, in the second case, the tracking of the x -component seems to have improved slightly, likely due to the decrease in mass.

From the velocity plots, it is possible to observe that, also in this case, the tracking of the x -component of the velocity appears to have improved slightly, both in terms of the time taken to reach the desired value and the overall performance, while the other component performs almost in the same way. Looking at the angular velocity plot, the tracking has improved across all components, with a decrease in magnitude as well. Lastly, looking at the z -component of the ground reaction force plots, it is possible to see that the magnitude decreases from 50 N to 25 N, while tracking remains almost unchanged.

In conclusion, the decrease in robot mass results in a slight improvement in velocity tracking across all components, along with a reduction in the overall magnitude.

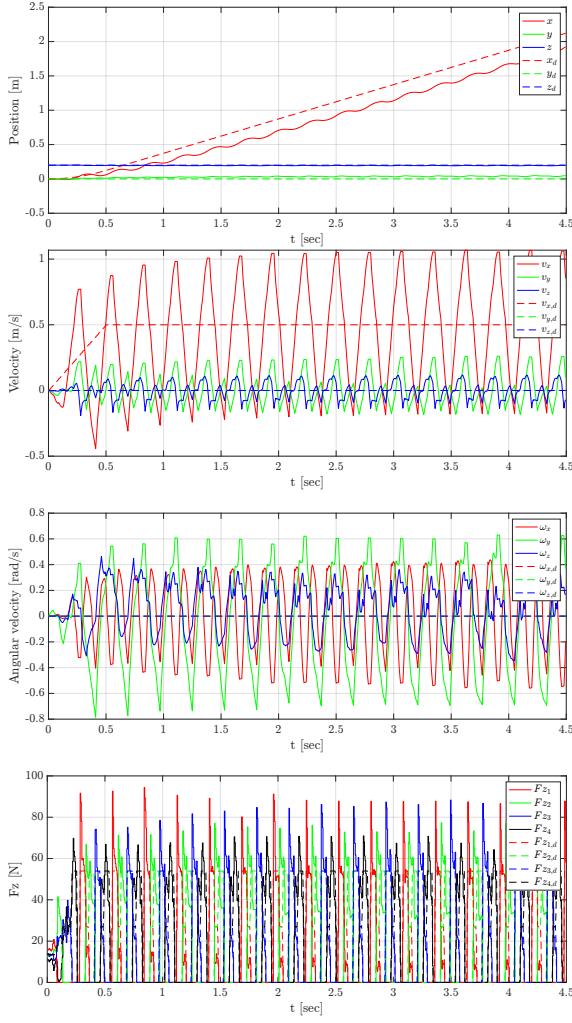


Figure 7: Gallop gait configuration results with $v_d = 0.5$ m/s, **mass** = 5.5 kg, $\mu = 1.0$

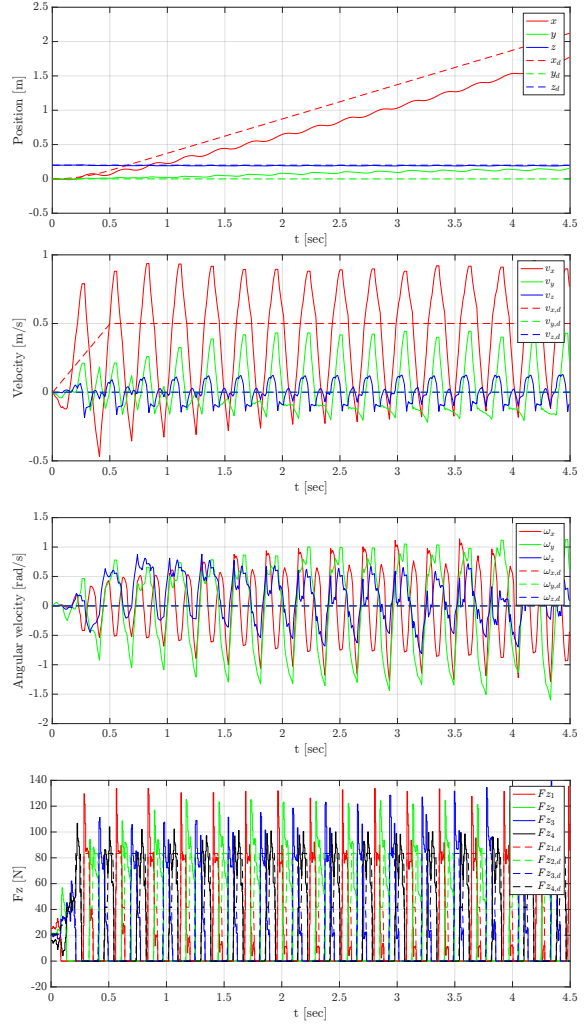


Figure 8: Gallop gait configuration results with $v_d = 0.5$ m/s, **mass** = 8.5 kg, $\mu = 1.0$

The fourth comparison regards the Gallop gait configuration. It is possible to see, on the left side (Figure 7), the configuration with the basic parameter values provided by the file, and on the right side (Figure 8), the configuration with adjusted parameters. Namely, the robot mass has been increased from 5.5 kg to 8.5 kg.

Looking at the position plots, it is possible to see that the tracking and behavior are similar in both cases. However, in the second case, the robot covers less distance—approximately 1.7 m compared to 2 m. This is due to the increase in mass.

From the velocity plots, it is possible to observe that the plots are quite similar, but the magnitude of the y -component of the velocity appears to have increased, while the other component performs almost in the same way.

Looking at the angular velocity plot, the magnitude has increased with respect to the basic parameters, due to the increase in mass.

Lastly, looking at the z -component of the ground reaction force plots, it is also possible to see that the magnitude increases from 100 N to 150 N, while the tracking remains almost unchanged.

In conclusion, the increase in robot mass results in similar tracking behavior, but a general increase in magnitude across the components.

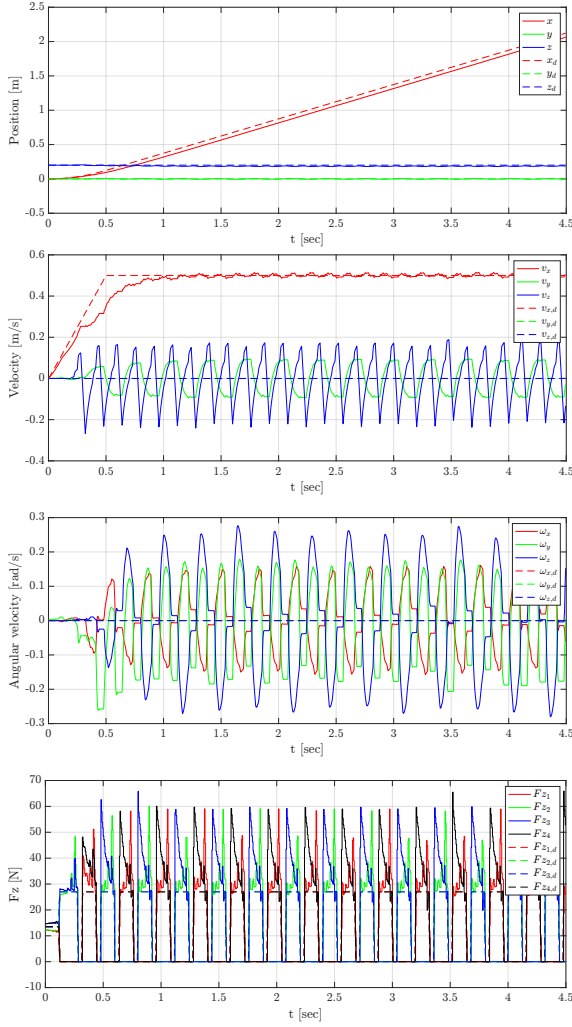


Figure 9: Trot run gait configuration results with $v_d = 0.5$ m/s, mass = 5.5 kg, $\mu = 1.0$

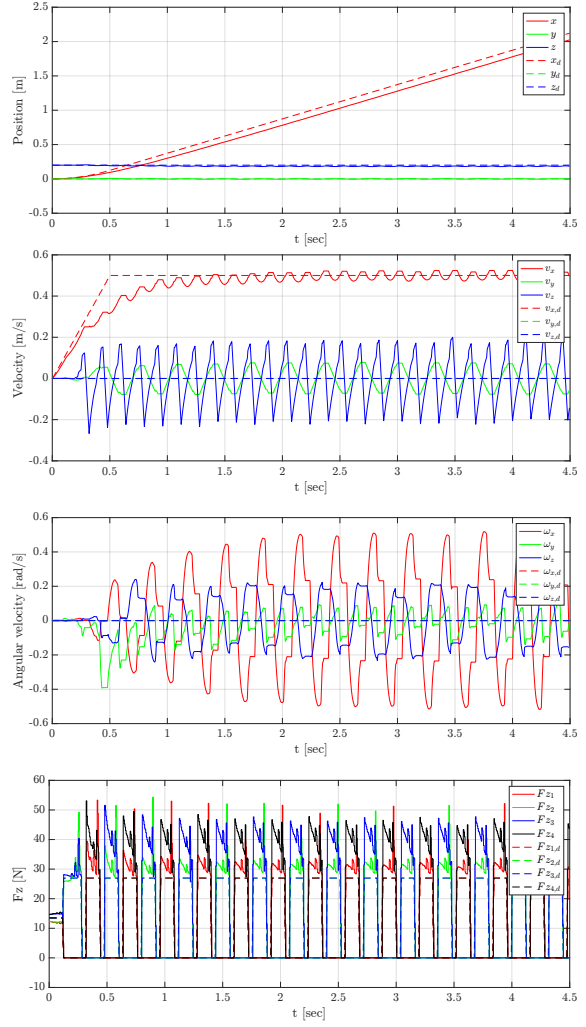


Figure 10: Trot run gait configuration results with $v_d = 0.5$ m/s, mass = 5.5 kg, $\mu = 0.1$

The fifth comparison regards the Trot run gait configuration. It is possible to see, on the left side (Figure 9), the configuration with the basic parameter values provided by the file, and on the right side (Figure 10), the configuration with adjusted parameters. Namely, the friction coefficient has been decreased from 1.0 to 0.1.

Looking at the position plots, it is possible to see that the tracking and behavior are quite similar in both cases.

From the velocity plots, it is possible to observe that the plots are quite similar, but the tracking of the x -component of the velocity occurs with less precision due to a more oscillatory trend, while the other component performs almost in the same way.

Looking at the angular velocity plot, the magnitude has increased with respect to the basic parameters, especially for the x -component of the angular velocity.

Lastly, looking at the z -component of the ground reaction force plots, it is also possible to see that the magnitude decreases from 80 N to 60 N, while the tracking remains almost unchanged. In conclusion, the decrease in the friction coefficient results in worse tracking behavior, but a reduction in magnitude across the components of the z -component of the ground reaction force.

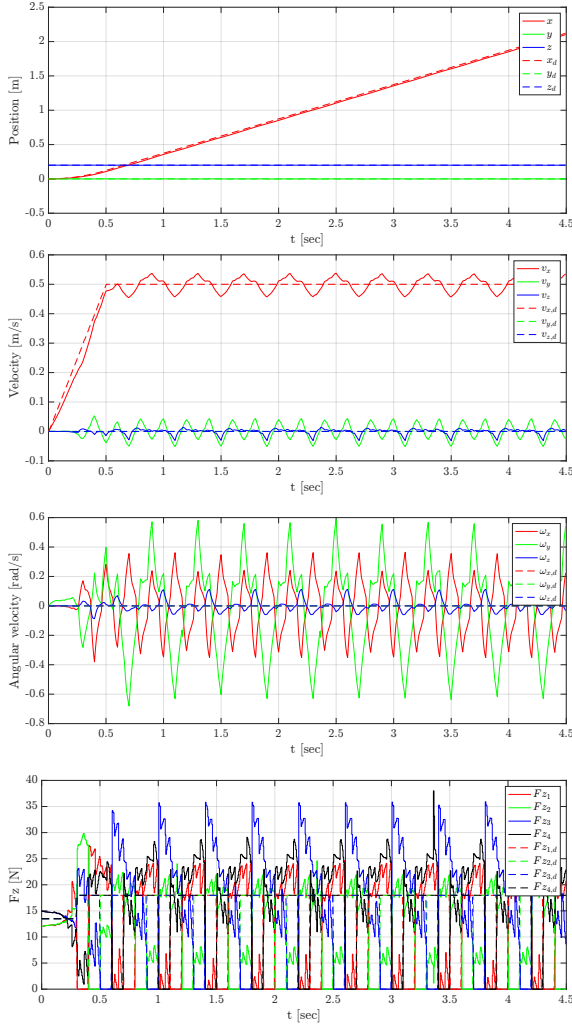


Figure 11: Crawl gait configuration results with $v_d = 0.5$ m/s, mass = 5.5 kg, $\mu = 1.0$

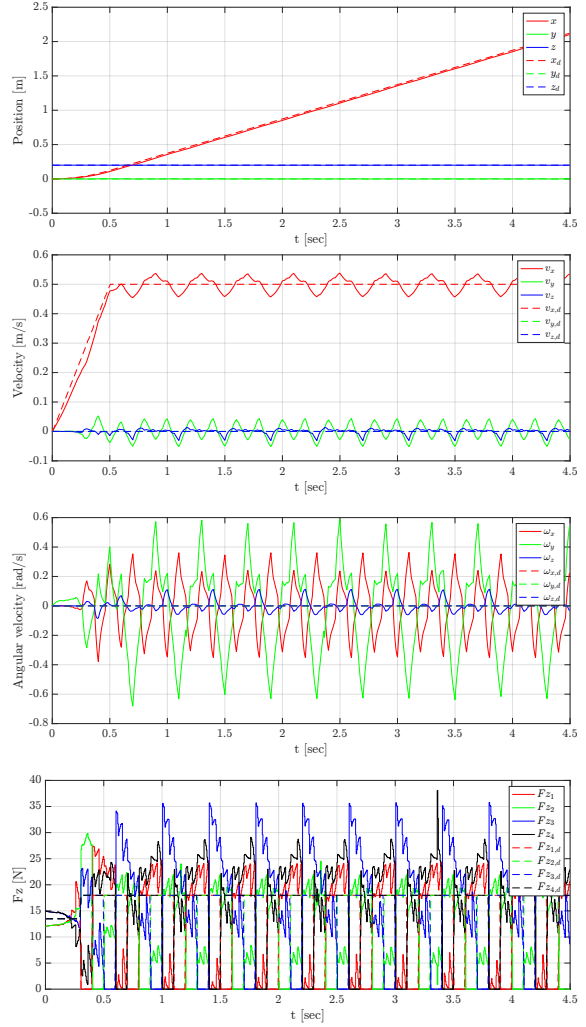


Figure 12: Crawl gait configuration results with $v_d = 0.5$ m/s, mass = 5.5 kg, $\mu = 2.0$

The last comparison regards the Crawl gait configuration. It is possible to see, on the left side (Figure 11), the configuration with the basic parameter values provided by the file, and on the right side (Figure 12), the configuration with adjusted parameters. Namely, the friction coefficient has been increased from 1.0 to 2.0.

Comparing the position, velocity, angular velocity, and the z -component of the ground reaction force plots, it is possible to see that the tracking and behavior are quite similar in both cases. This was to be expected, as increasing the friction coefficient beyond the unit value does not lead to any improvement or deterioration in terms of performance.

In conclusion, the increase in the friction coefficient does not affect the robot's performance.

In summary, the overall analysis can be structured with respect to the three parameters considered: desired velocity, robot mass, and friction coefficient.

If increasing the desired velocity, as can be seen in the Bound gait configuration, there is a slight improvement in velocity tracking for the x -component. If decreasing the desired velocity, as observed in the Trot gait configuration, there is no significant effect in terms of improving or worsening the overall performance.

For what concerns the robot mass, if increasing it, like in the Gallop gait configuration, the tracking behavior remains similar, but there is a general increase in magnitude across the components. If decreasing the mass, as in the Pacing gait configuration, there is a slight improvement in velocity tracking across all components, along with a reduction in the overall magnitude.

In conclusion, regarding the friction coefficient, if decreasing it, as shown in the Trot run gait configuration, the result is a worse tracking behavior, but also a reduction in magnitude across the components of the z -component of the ground reaction force. If increasing it, as in the Crawl gait configuration, there is no effect on the robot's performance.

4 Point

Starting from the MATLAB file `rimless_wheel.m`, which simulates the motion of a rimless wheel, it is possible to perform an analysis by changing just the initial angular velocity.

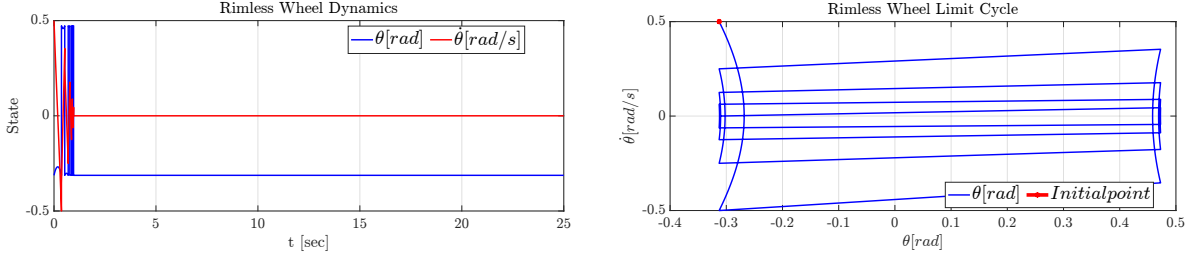


Figure 13: $\dot{\theta}(0^+) = 0.5 \text{ rad/s}$, $l = 1$, $\alpha = \pi/8$, $\gamma = 0.08 \text{ rad}$

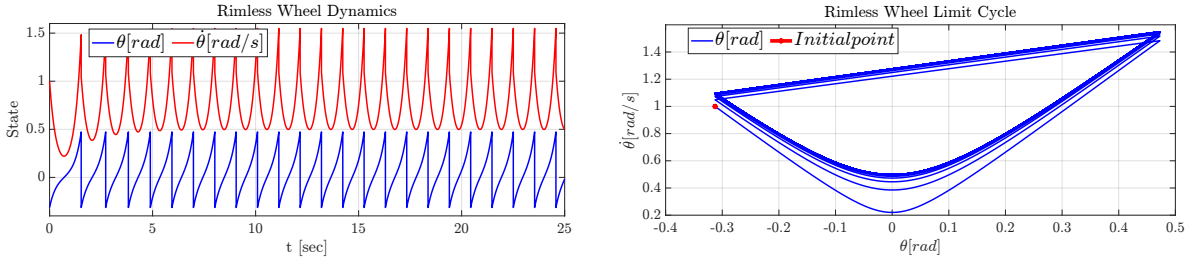


Figure 14: $\dot{\theta}(0^+) = 1 \text{ rad/s}$, $l = 1$, $\alpha = \pi/8$, $\gamma = 0.08 \text{ rad}$

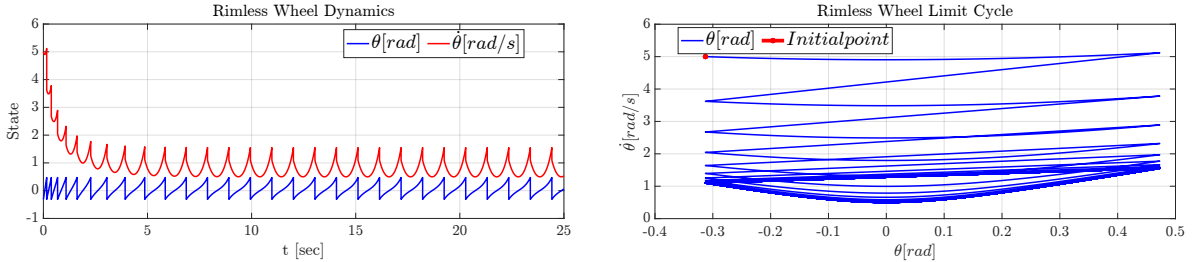


Figure 15: $\dot{\theta}(0^+) = 5 \text{ rad/s}$, $l = 1$, $\alpha = \pi/8$, $\gamma = 0.08 \text{ rad}$

In Figure 13, there is the Rimless Wheel with initial angular velocity $\dot{\theta}(0^+) = 0.5 \text{ rad/s}$, where it is possible to see the time histories of the states and the resulting phase portrait. These plots suggest the initial conditions do not lie within the basin of attraction of a limit cycle. Consequently, the rimless wheel is not able to maintain dynamic walking after about 1 s and stops asymptotically, with no limit cycle.

This behavior is reasonable because forward walking occurs when the system has an initial velocity $\dot{\theta}(0^+) > \omega_1$, where $\omega_1 = \sqrt{2\frac{g}{l}(1 - \cos(\gamma - \alpha))} = 0.9754$ is the threshold above which the system is able to take a step and so with this configuration the wheel does not have enough initial velocity to perform the passivity walk.

In Figures 14 and 15, two different configurations for the initial angular velocity are presented, respectively $\dot{\theta}(0^+) = 1 \text{ rad/s}$ and $\dot{\theta}(0^+) = 5 \text{ rad/s}$. Both satisfy the threshold condition and in both cases, the system converges to a stable limit cycle, although the convergence occurs more rapidly in the first case, while in the second case it takes approximately 5 s to stabilize.

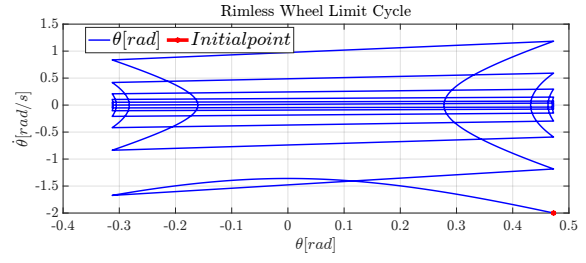
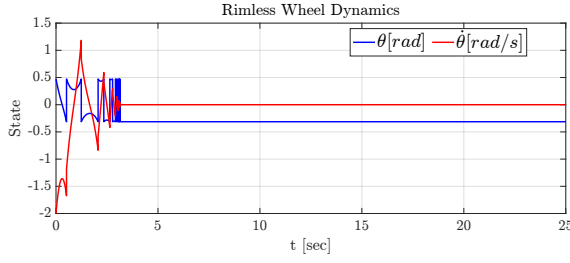


Figure 16: $\dot{\theta}(0^+) = -2\text{rad/s}$, $l = 1$, $\alpha = \pi/8$, $\gamma = 0.08\text{rad}$

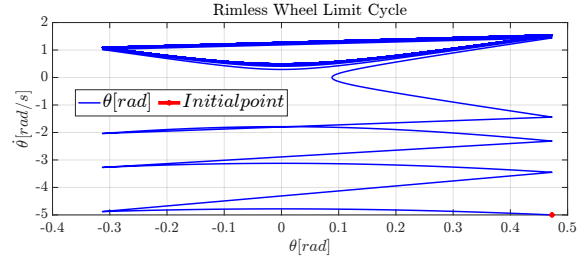
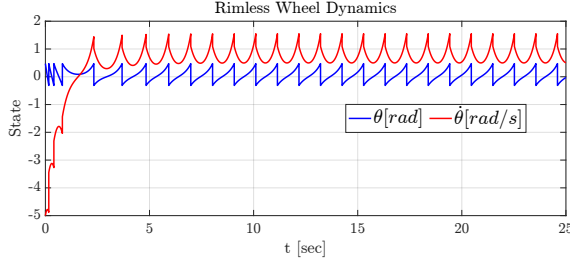


Figure 17: $\dot{\theta}(0^+) = -5\text{rad/s}$, $l = 1$, $\alpha = \pi/8$, $\gamma = 0.08\text{rad}$

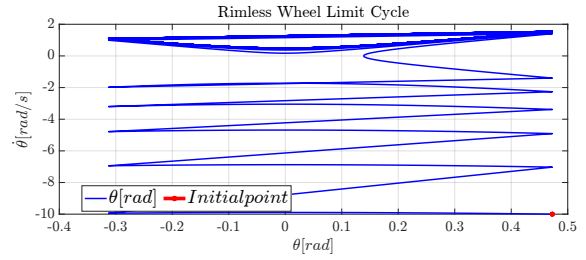
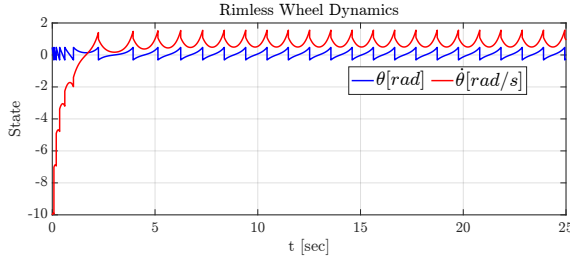


Figure 18: $\dot{\theta}(0^+) = -10\text{rad/s}$, $l = 1$, $\alpha = \pi/8$, $\gamma = 0.08\text{rad}$

From the phase portraits, it is evident that both initial conditions lie within the basin of attraction of the same limit cycle. However, due to the different initial angular velocities, the system in Figure 14 converges to the limit cycle more rapidly than in Figure 15. Other tests have confirmed that the size of the basin of attraction is sufficiently large to allow convergence to the limit cycle even for values of $\dot{\theta}(0^+)$ greater than those reported.

The same analysis can be done to negative values of the initial angular velocity. In Figure 16, it is possible to see that for a small negative initial velocity $\dot{\theta}(0^+) = -2\text{rad/s}$, the system is not able to maintain dynamic walking after about 3s and stops asymptotically, with no convergence to limit cycle.

In Figure 17 and Figure 18 it is possible to see that for larger (more negative) initial velocities $\dot{\theta}(0^+) = -5\text{rad/s}$ and $\dot{\theta}(0^+) = -10\text{rad/s}$, the system converges to a stable limit cycle after about 3s. These results suggest that the corresponding basins of attraction are similar to those observed in the positive velocity cases.

On the other hand, it is possible to pursue an analysis modifying the leg length, the inter-leg angle, and the slope inclination provided by the code, while keeping the initial angular velocity fixed at 0.95rad/s .

In order to evaluate the effects provided by the single changing of parameters, it has been chosen to change only one parameter for time (by increasing and decreasing it), while keeping the original parameters provided by the file, and then see if the system still converges to a limit cycle and how the characteristics of the limit cycle may change.

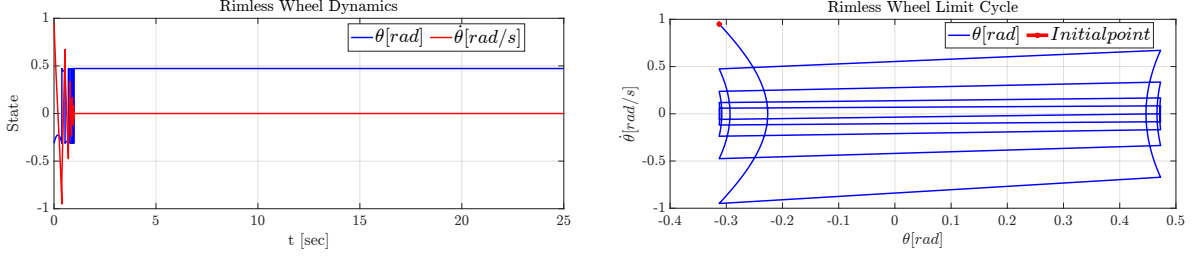


Figure 19: $\dot{\theta}(0^+) = 0.95\text{rad/s}$, $l = 0.5\text{m}$, $\alpha = \pi/8$, $\gamma = 0.08\text{rad}$

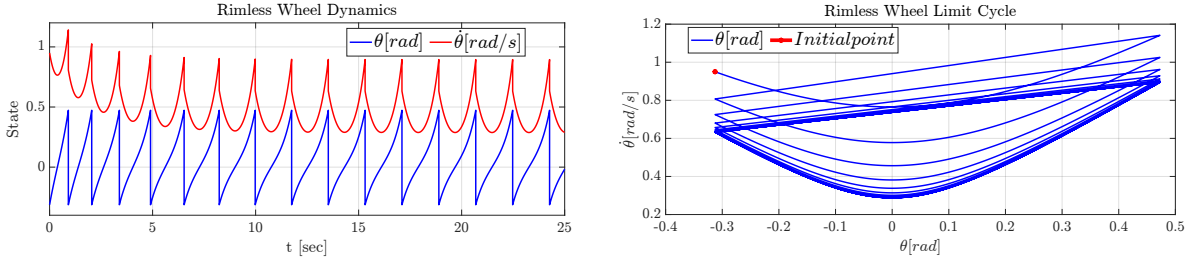


Figure 20: $\dot{\theta}(0^+) = 0.95\text{rad/s}$, $l = 3\text{m}$, $\alpha = \pi/8$, $\gamma = 0.08\text{rad}$

First, the effect provided by the changing of the leg length parameter l is considered, with two test cases.

In Figure 19, a reduced leg length $l = 0.5\text{m}$ results in the wheel stop after about 0.5s , indicating insufficient energy to sustain motion and thus no limit cycle is observed.

In contrast, in Figure 20 it is possible to see that increasing the leg length to $l = 3\text{m}$ allows the system to converge to a stable limit cycle after approximately 5s . The phase portrait shows that the angular velocity oscillates within the range $0.3\text{--}0.9\text{rad/s}$, confirming the presence of dynamic walking.

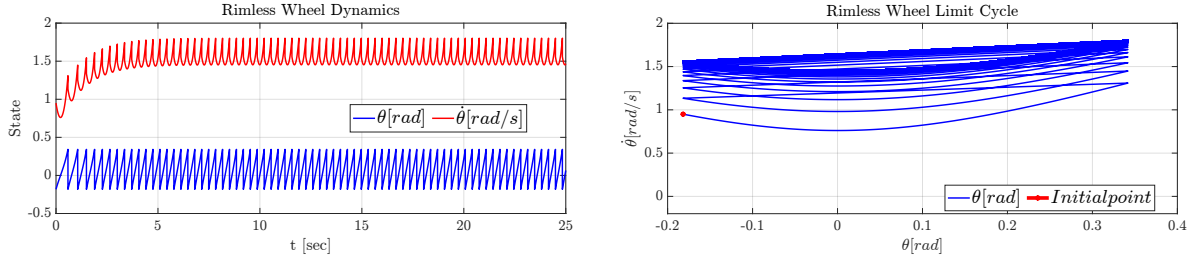


Figure 21: $\dot{\theta}(0^+) = 0.95 \text{ rad/s}$, $l = 1 \text{ m}$, $\alpha = \pi/12$, $\gamma = 0.08 \text{ rad}$

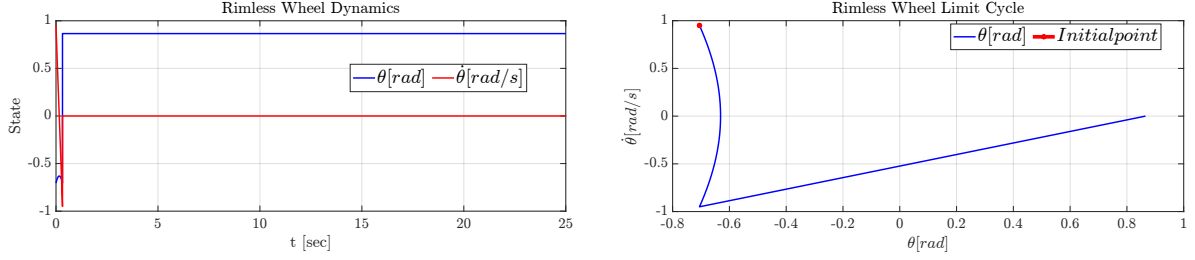


Figure 22: $\dot{\theta}(0^+) = 0.95 \text{ rad/s}$, $l = 1 \text{ m}$, $\alpha = \pi/4$, $\gamma = 0.08 \text{ rad}$

With regard to the inter-leg angle parameter α , two cases have been considered.

In Figure 21 it is possible to see that for a smaller angle $\alpha = \pi/12$, the system converges to a stable limit cycle after approximately 5 s. The angular velocity oscillates within the range 1.2–1.7 rad/s, indicating successful passive walking.

In Figure 22 it is possible to see that with a larger angle $\alpha = \pi/4$, the wheel is not able to sustain motion and comes to rest after about 1 s, showing no convergence to a limit cycle.

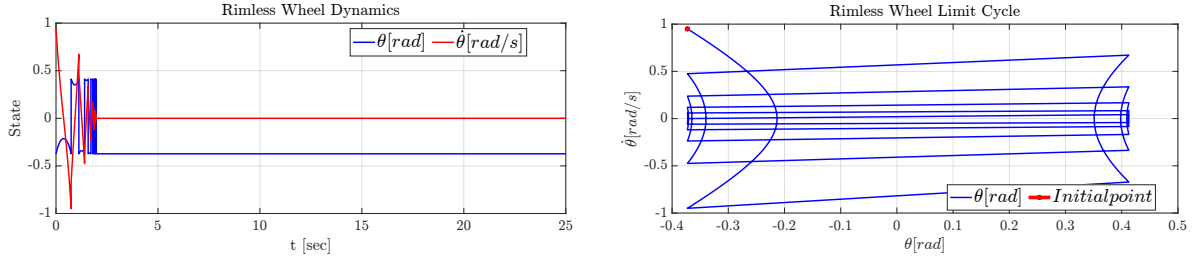


Figure 23: $\dot{\theta}(0^+) = 0.95 \text{ rad/s}$, $l = 1 \text{ m}$, $\alpha = \pi/8$, $\gamma = \mathbf{0.02 \text{ rad}}$

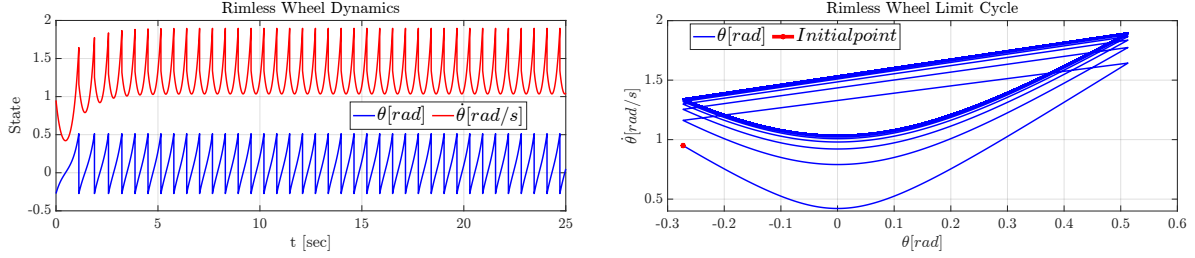


Figure 24: $\dot{\theta}(0^+) = 0.95 \text{ rad/s}$, $l = 1 \text{ m}$, $\alpha = \pi/8$, $\gamma = \mathbf{0.12 \text{ rad}}$

Finally, with regard to the slope inclination parameter γ , two cases have been taken into account.

In Figure 23 it is possible to see that for a lower inclination $\gamma = 0.02 \text{ rad}$, the system does not have sufficient energy to sustain passive motion and comes to rest after approximately $2/3 \text{ s}$. As a result, no limit cycle is observed.

In Figure 24 shows the behavior for a steeper slope $\gamma = 0.12 \text{ rad}$, where the system quickly converges to a stable limit cycle after about $2/3 \text{ s}$. The angular velocity oscillates in the range $1\text{--}1.8 \text{ rad/s}$, indicating sustained dynamic walking.

In conclusion, based on the chosen analysis criteria, it can be said that the parameters whose variations significantly affect the convergence to a limit cycle are the decrease in leg length and slope inclination, as well as the increase in the inter-leg angle.

Regarding the resulting limit cycles, it has been observed that these parameters lead to huge changes in the shape and position of the limit cycles in the phase plane. In particular, both the trajectory location and the oscillation range of the limit cycles are affected by the specific parameter being varied. However, when only the initial angular velocity was changed, the shape and range of the limit cycles remained nearly unchanged.